SOP-QA-014 | Controlled Document Lifecycle

Version 1.0 • Status: DRAFT • Effective date: NOT ASSIGNED • Process owner: Quality Systems

TRAINING EXAMPLE ONLY — Fictional organization, synthetic records, and invented personnel. Not a validated procedure, production instruction, executed approval, or assertion of regulatory compliance.

Document control record

Field	Entry
Document ID / version	SOP-QA-014 / 1.0
Document class	Standard Operating Procedure
Process owner	Quality Systems Manager (fictional)
Approver	Designated Quality approver (not signed)
Change request	DCR-2026-007 (open synthetic example)
Supersedes	None — sample initial issue
Effective date	Not assigned until authorized approval
Review schedule	Annual in this hypothetical company policy; change-triggered review sooner
Applicability	Fictional GxP-relevant SOPs, controlled work instructions, forms and templates

1. Purpose

Define a repeatable, traceable method for authoring, checking, reviewing, approving, publishing, training on, periodically reviewing, revising, and retiring controlled documents. The process is designed to prevent unintended use of obsolete procedures and preserve review history and training evidence.

2. Scope

Applies to documents describing regulated material receipt, QC testing, Manufacturing, product disposition, equipment control, deviation/CAPA, supplier oversight, and supporting quality processes. It does not establish manufacturing specifications, clinical trial procedures, or clinical-site work instructions. If a required site-specific quality agreement or approved procedure differs, the authorized system owner must resolve the conflict before implementation.

3. Terms and definitions

Term	Meaning
Controlled document	Approved, versioned procedural content subject to defined access and change control.
Document owner	Accountable SME/process owner who confirms technical accuracy and suitability.
Document control	Function that administers ID, metadata, routing, approved publication, archival and version integrity.
Effective version	Formally approved and released version authorized for use on an assigned effective date.
Training impact assessment	Evaluation of who needs training and how competence/acknowledgment is evidenced before work.
Superseded	Earlier approved document retired from routine use but

	retained under applicable record rules.
DCR	Document change request: tracks reason, impact, reviewers and disposition.

4. Roles and responsibilities

Role	Accountability
Requestor / originating SME	Propose change; cite observations and affected documents; identify stakeholders.
Document author	Draft clear task-oriented steps; check terminology, forms and exceptions; maintain change summary.
Document Control	Assign identifier; confirm correct template; check links, metadata, version and approval route; publish approved revision.
Quality Assurance	Evaluate GMP/quality risk, unresolved comments, deviations, approvals and release eligibility.
Quality Control	Verify references to test methods, sample data, laboratory records and acceptance criteria when affected.
Manufacturing / Supply Chain	Review operational feasibility, receiving and supplier-data handoffs where applicable.
Training Coordinator	Assign role-based training, verify completion and keep records; escalate overdue tasks.
Document users	Use only the effective version and promptly report conflicting or inaccurate instructions.

5. Prerequisites and inputs

- Approved document template and current controlled glossary; applicable SOP and form references.
- Document-change request or new-document request with source issue, reason, and planned implementation.
- Relevant quality-event references, risk review, affected departments, records and training requirements.
- Access to authorized document repository, version history, and approved approval workflow.

6. Detailed workflow

6.1 Initiate and classify

Requestor records document ID or proposed title, affected process and sections, reason for change, relevant quality event, intended users, urgency, and whether a linked material/vendor change is in scope. Document Control assigns a DCR ID and identifies the owner.

6.2 Assess process and quality impact

Owner/QA identify related SOPs, forms, specifications, laboratory data, material statuses, supplier approvals, training roles, validation considerations, and any regulatory filing impact; engage QC/Manufacturing/Procurement where relevant. Record rationale for items deemed not applicable.

6.3 Prepare draft

Author uses sequential instructions, named roles, expected system or record inputs, decision points, explicit exceptions, and associated evidence. Use controlled terminology; distinguish example values from acceptance criteria. Mark draft prominently and use a non-effective version identifier.

6.4 Perform editorial QC

Document Control verifies title/ID/version, definitions, navigation, cross-references, document-control fields, form links, completeness of steps, verb consistency, accessibility and readability. Resolve broken links and conflicting numbers.

6.5 Conduct SME and cross-functional review

Route to affected SMEs, including QC for laboratory requirements and Supply Chain for vendor/material records. Reviewers compare instructions with approved governing procedures, current system fields, evidence and handoffs. Record each comment and disposition; reopen unresolved safety/compliance issues.

6.6 Complete QA review and approval

QA evaluates residual risk, correctness of references, linked deviations/CAPA, changes to specifications, adequacy of controls, and approved implementation plan. Authorized approvers sign only in the designated system. A submitted request or email discussion does not constitute approval.

6.7 Publish effective version

After required signatures, Document Control records the authorized revision and effective date, restricts editable master access, updates document register and ensures only the effective revision is distributed at the point of use. Archive superseded copies without erasing their history.

6.8 Assign and verify training

Training Coordinator maps affected roles, assigns the designated training method, communicates due dates, and records acknowledgment or competency checks as required. For work impacted by a revision, verify required training before personnel conduct that work.

6.9 Monitor periodic and triggered review

Owner reviews by company-defined schedule or sooner after process, system, supplier, regulatory or quality-event changes. Record no-change confirmations or open a new DCR. Overdue reviews must be risk-triaged; do not automatically presume an expired document is safe or unsafe.

6.10 Retire / supersede

Document Control records replacement ID, retirement reason, owner approval and retention rule; withdraws obsolete point-of-use copies and preserves searchable audit trails.

7. Decision and escalation rules

Condition	Required action	Evidence
Missing required approval	Hold publication; notify QA and Document Control.	Approval routing and open-item log
Conflicting source instruction / QC record	Pause change; obtain SME/QC clarification; assess whether deviation is required.	Comment disposition and event reference
Unresolved material/vendor record inconsistency	Escalate to QA, Procurement and Receiving; prevent unauthorized material use under applicable approved process.	Material status check and issue record
Required training overdue	Escalate to owner/Training before performing affected task.	Training completion / exception record
Obsolete file at point of use	Remove or restrict; assess potential impact through quality-event process.	Access control and event disposition

8. Required records and retention

Maintain DCR, document review comments, risk/impact review, approval audit trail, effective-version register, archived superseded revision, training assignment/completion history, and periodic-review record according to approved company retention procedures. This portfolio deliberately does not prescribe a universal retention period.

9. Quality and usability checks

- Each procedure step has a clear responsible role, trigger, action and required record.
- No draft is represented as approved or effective; invented signatures are not used.
- All forms and SOP references use matching IDs and current revision references.
- A reviewer unfamiliar with the process can follow the normal path and locate escalation conditions.
- Quality and training handoffs can be reconstructed from the sample evidence trail.

10. References and interpretation

- 21 CFR Part 211 — CGMP for finished pharmaceuticals; relevant provisions include quality-control responsibilities, written procedures, and records. Applicability depends on actual product/process.
- 21 CFR Part 11 — criteria for certain FDA-regulated electronic records and electronic signatures when within scope; not a general guarantee of software validation.
- ICH Q10 — pharmaceutical quality system model and lifecycle approach to change management and CAPA; guidance, not a claim of facility certification.
- ICH Q9(R1) — quality risk management principles relevant to risk-based evaluations.
- ICH E6(R3) concerns clinical-trial GCP; it is not an applicable manufacturing SOP standard for this fictional incoming-materials workflow. No human clinical-site operations are represented.

11. Revision history

Version	Date	Author	Summary	Status
1.0	2026-11-02	Portfolio author	Initial detailed controlled document workflow sample	DRAFT — not approved

Appendix A. Publication readiness control

This reviewer aid is a proposed verification template, not proof of completed review. A production organization would configure required checkboxes and signatures in its authorized document-management system.

Review point	Required verification	Example status
Technical content	Affected SMEs confirm tasks, forms, system fields and SOP references.	NOT REVIEWED
Cross-functional impact	QC, Manufacturing, Procurement and QA inputs addressed if applicable.	NOT REVIEWED
Controlled metadata	ID, version, owner, approved signers, links and change history checked.	NOT REVIEWED
Quality event traceability	Deviation, CAPA and change-control IDs linked and open actions risk-assessed.	NOT REVIEWED
Training	Affected roles, modality and pre-use training requirement identified.	NOT REVIEWED
Distribution	Approved master published and superseded point-of-use copies withdrawn.	NOT PUBLISHED
Final authority	Required approvals recorded by authorized staff in designated system.	NOT APPROVED

REGULATORY / TECHNICAL REVIEW ADDENDUM — November 2026

Independent technical writing portfolio enhancement. Additional validation gates and examples are illustrative, not authorized production instructions. Regulated applicability varies by product, process, record and organization.

Regulatory scope, approval authority, and document status

This controlled-style SOP is a demonstration: no effective date, approval signature, production authority or actual training record is issued. The company quality unit—not the portfolio author—must decide applicable predicate regulations, approve production-impacting procedures, and authorize implementation. 21 CFR 211.22(c)-(d) places approval/rejection of quality-impacting procedures and written quality-unit responsibilities with the quality control unit.

Separate four states: DRAFT (editable, not in use), IN REVIEW (controlled review routing), EFFECTIVE (authorized version and trained users), and SUPERSEDED/RETIRED (restricted from point of use, retained under policy). A completed draft or mock signature is not an electronic signature under Part 11.

Change-impact assessment: worked example

Example DCR-2026-007 requests revision of a material receipt checklist because the vendor master contains outdated supplier IDs. Document Control records current document rev D, proposed rev E, linked exception EX-2026-018, affected systems (ERP/material register/QMS), material families, site owners, and the first planned training group.

QA checks whether revised instructions affect component acceptance or product quality; QC checks whether sampling or testing references need change; Procurement confirms vendor master ownership; Manufacturing checks point-of-use consequences; IT/CSV assesses whether a regulated electronic workflow is changed. Unresolved material disposition stays on hold until authorized quality review.

Sample acceptance criterion: every change-control record must link an approved revision, impact assessment, reviewer comments, training decision, effective date and documented disposition of prior versions. This is a proposed company criterion, not a universal statutory timeline.

Good documentation practice and record correction

Write activities contemporaneously, clearly and attributable to the person performing the work. Use actual dates and references; do not backdate; preserve original values and the reason for corrections. In paper workflows, the site SOP should prescribe legible permanent corrections without obliteration. In validated electronic workflows, corrections must preserve an appropriate history under applicable predicate requirements and system controls.

The familiar ALCOA/ALCOA+ terms are data-integrity principles drawn from industry practice and FDA data-integrity guidance; they do not create a separate FDA certification. Use them as review prompts for attribution, legibility, contemporaneity, original/accurate records, completeness, consistency, endurance and availability.

Document release checklist: evidence and ownership

DCR-2026-007: Owner documents the reason and scope; QA signs off on quality impact; QC acknowledges testing/sampling implications; Supply Chain confirms supplier-record instructions; Training assigns role-specific learning; Document Control checks document ID/revision/cross-references before publication.

Potential failures and response: missing reviewer approval -> return to review; unanswered comment -> hold approval; missing training-impact disposition -> block release of training-dependent work; superseded copy in a workstation -> withdraw and document correction; broken referenced form -> repair before effective publication.

Document-specific reference basis and status

FDA 21 CFR 211.22 — quality unit authority (full URL in portfolio review report)

FDA 21 CFR 211.84 — component sampling/testing/release (full URL in portfolio review report)

FDA 21 CFR Part 11 — electronic records and e-signatures (full URL in portfolio review report)

FDA Part 11 scope and application guidance (full URL in portfolio review report)

FDA Data Integrity and Compliance With Drug CGMP (2018) (full URL in portfolio review report)

ICH Q9(R1) Quality Risk Management (2023) (full URL in portfolio review report)

ICH Q10 Pharmaceutical Quality System (full URL in portfolio review report)

ICH Q8/Q9/Q10 Q&A R5 (May 2026) (full URL in portfolio review report)

Status: technical/editorial review against publicly available references completed at a portfolio-example level; no licensed-standard full-text conformance audit, live environment testing, site QA approval or legal certification performed.

Appendix: November 2026 standards and documentation applicability review

PUBLIC SYNTHETIC PORTFOLIO SAMPLE — This appendix records source-checking and editorial scope boundaries. It does not make the fictional organization regulated, certify a system, or replace actual company policies and competent quality/legal review.

Pharmaceutical regulatory scope and data-integrity editorial verification

- 21 CFR 211.22: fictional QA unit responsibility is distinct from QC testing and procurement approval; incoming component disposition must not be inferred from data-field reconciliation alone.
- 21 CFR 211.84: describe quarantine, identity testing and QC release expectations only to the extent that the hypothetical material falls under the rule; signed review of a supplier certificate is not automatic approval.
- 21 CFR 211.180/211.192 and FDA data-integrity guidance: reconcile original records, trace corrections, record investigations, preserve legibility and demonstrate review of unexplained discrepancies.
- 21 CFR Part 11: evaluate record and electronic-signature predicate-rule scope; do not claim that every spreadsheet or SOP falls under Part 11.
- ICH Q9(R1)/Q10 and FDA May 2026 Q8/Q9/Q10 Q&A: risk-based investigation, change control, oversight, effectiveness checks and management review are guidance/model concepts, not blanket product approvals.
- 2026 FDA QMSR/21 CFR 820 and February 2026 Computer Software Assurance: relevant to in-scope medical-device workflows; not universally required for this hypothetical drug-component process.
- Document design check: explicit document ID/version/status, role-based responsibilities, objective acceptance criteria, accurate dates, no invented e-signatures or final QA approvals, and clear synthetic label.

Reference links and recheck notes

https://www.ecfr.gov/current/title-21/chapter-I/subchapter-C/part-211

https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-11

https://www.fda.gov/regulatory-information/search-fda-guidance-documents/data-integrity-and-compliance-drug-cgmp-questions-and-answers-guidance-industry

https://www.fda.gov/medical-devices/postmarket-requirements-devices/quality-management-system-regulation-qmsr

Verified against published authoritative source pages as of November 2, 2026. No proprietary standards were audited clause-by-clause. Actual application must be assessed based on product, system use, region, and record type. Proposed procedures have not been executed in a production environment.

Independent second-pass document review | November 2, 2026

REVIEWED PORTFOLIO DEMONSTRATION — this appendix records a document-specific editorial and source-scope check. The main procedure and forms are fictional and not approved for production, executed testing or regulated use. Original source versions are retained as editable files in this portfolio.

A. Document-specific findings and checks

Review objective	2026 reviewer correction / expectation	Verification evidence
Control SOP scope	Define author, SME reviewer, independent quality-unit approval, effective date, superseded status, controlled-copy release and periodic review.	Document register has approved version, owner, effective status, training link and retirement evidence.
Training / change	Determine risk-driven retraining scope and record completion before relevant work under revised procedure.	Show training impact, approved training assignment and document release linkage.
Data integrity	Record complete, contemporaneous document approvals and maintain audit trail for controlled electronic documents where applicable.	Ensure no fictional signoff is shown as actual approval; examples must show DEMONSTRATION / NOT EXECUTED.

B. Applicable regulatory / guidance boundaries

Review objective	2026 reviewer correction / expectation	Verification evidence
Drug CGMP / Quality unit	21 CFR 211.22, 211.80, 211.84, 211.100, 211.160 as relevant	Mandatory only when the fictional activities actually fall within covered finished-drug operations; paperwork precheck is not QC material disposition.
Records integrity / electronic signature	21 CFR 211.180, 211.188, 211.194; Part 11 conditional	Trace data to author, time, source and approved version; do not assume that every spreadsheet is a Part 11 record.
Risk and change	ICH Q9(R1), ICH Q10; FDA 2018 final data integrity guidance	Nonbinding guidance/system model as adopted or implemented; distinguish a risk-based recommendation from regulation.

C. Evidence and sign-off boundaries

Status: Editorial second-pass reviewed; technical procedure not executed; company-specific SOP approval not obtained.

All tests, screenshots, users, suppliers, dates, IDs, incident observations, performance metrics and signatures in the demonstration are synthetic.

Next practitioner action: confirm source applicability for the actual jurisdiction and product; conduct SME/user testing and accessibility review; update controlled revision and approval fields.

No color, watermark, revision number, PDF export or reference link by itself demonstrates regulatory compliance.

D. Authoritative primary-source references

FDA drug CGMP 21 CFR Part 211: https://www.ecfr.gov/current/title-21/chapter-I/subchapter-C/part-211

FDA 21 CFR Part 11 scope: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/part-11-electronic-records-electronic-signatures-scope-and-application

FDA drug CGMP final data integrity: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/data-integrity-and-compliance-drug-cgmp-questions-and-answers

ICH Q9(R1)/Q10: https://admin.ich.org/page/quality-guidelines

FDA device QMSR Feb 2026: https://www.fda.gov/medical-devices/postmarket-requirements-devices/quality-management-system-regulation-qmsr